Step Insertions of Reactivity

NE 630 | Nuclear Reactor Theory

28
LESSON

Read: FNRP 5.5.

Students will be able to solve the kinetics equations subject to step insertions of reactivity and to predict neutron and precursor densities and reactor periods from those solutions and approximations thereof.

1 Review: The Point Kinetics Equations

The time evolution of the neutron population (or density!) is described by

$$\frac{dn}{dt} = \left(\frac{\rho(t)-\beta}{\Lambda}\right) n(t) + \sum_{i=1}^{N} \lambda_i C_i(t) + S(t)\,, \qquad \text{(FNRP 5.47)}$$

while the precursor populations (or densities!) evolve according to

$$\frac{dC_i}{dt} = -\lambda_i C_i(t) + \frac{\beta_i}{\Lambda} n(t) \quad i = 1, 2, \ldots, N\,, \qquad \text{(FNRP 5.48)}$$

where $\beta = \sum_i \beta_i$ is the total delayed neutron precursor fraction. These delayed neutrons increase the average neutron lifetime[1] according to

$$\bar{l} = (1-\beta) l + \beta l_d \qquad \text{(FNRP 5.36)}$$
$$= l + \beta/\lambda\,.$$

2 Review: A Single Precursor Group

One can reduce the N precursor equations to one by defining

$$\beta = \sum_{i=1}^{N} \beta_i \quad \text{and} \quad \frac{1}{\lambda} = \left(\frac{1}{\beta}\sum_{i=1}^{N}\frac{\beta_i}{\lambda_i}\right). \qquad \text{(FNRP 5.31 \& 5.34)}$$

Then the PKEs reduce to

$$\frac{dn}{dt} = \frac{\rho(t)-\beta}{\Lambda} n(t) + \lambda C(t) \quad \text{and} \quad \frac{dC}{dt} = -\lambda C(t) + \frac{\beta}{\Lambda} n(t)\,.$$

Example: Assume a ^{235}U-fueled reactor is initially critical, i.e., $\rho(t<0) = 0$, with $n(0) = 10^9$. With $\Lambda = 10^{-4}$ and $\rho(t \geq 0) = \rho_0 = 0.25\$$, write down the equations, with numerical coefficients, needed to find $n(t)$ and $C(t)$.

Ans. Using data from Table 5.1, $\beta = 0.00650$ and $\lambda = 0.07664$. Then, FNRP (5.29) gives $C(0) = \beta n(0)/(\lambda\Lambda) \approx 8.48 \cdot 10^{11}$. Finally, $\rho_0 = 0.25\ (\$) = 0.25\beta\ (\Delta k/k) = 162.5$ (pcm).

3 Step Insertions and the In-Hour Equation

Whether we have N precursor groups or just one group, the PKEs can be solved analytically only for special cases in which $\rho(t)$ and $S(t)$ are simple functions of time (e.g., a constant or a sinusoid).

For a constant, **step insertion of reactivity** $\rho(t) = \rho$, the PKEs have *constant coefficients*.

If we further take $S(t) = 0$, then the solutions (i.e., $n(t)$, $C_i(t)$) must be sums of exponential functions with different time constants.

In other words,

$$n(t) = \sum_{j=1}^{N+1} A_j e^{\omega_j t}$$

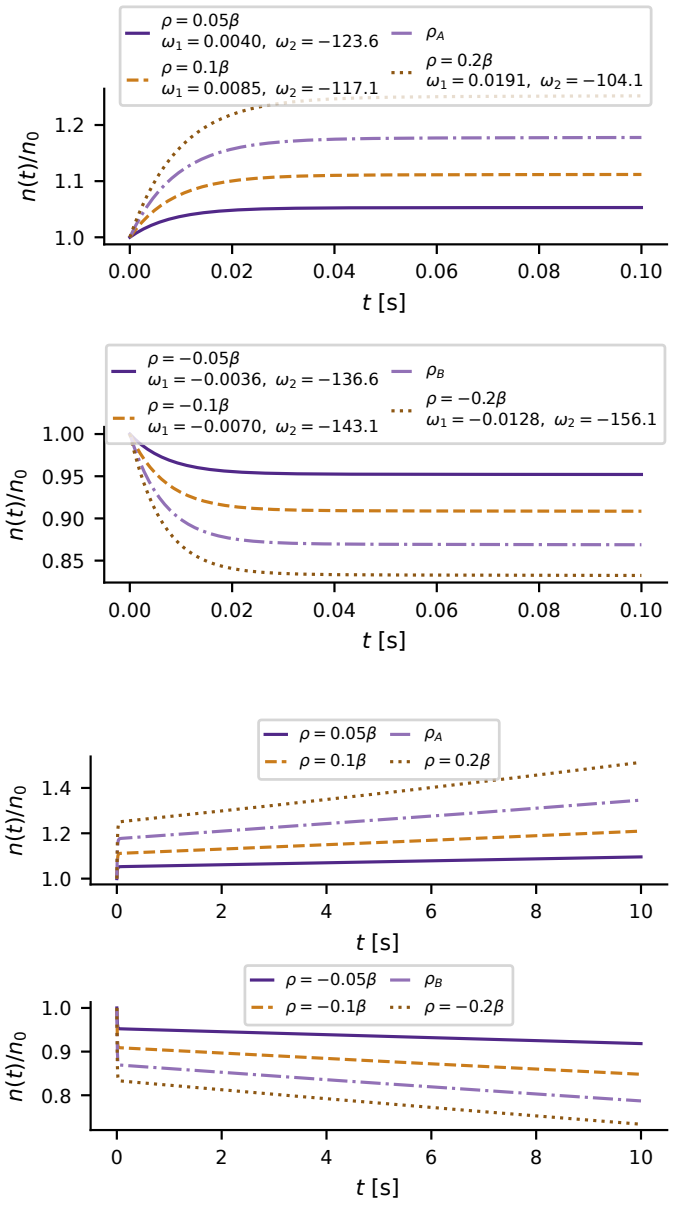
$$C_i(t) = \sum_{j=1}^{N+1} B_{ij} e^{\omega_j t}\,,\ i = 1, \ldots, N\,,$$

whose time constants ω_j satisfy

$$\boxed{\rho = \left(\Lambda + \sum_{i=1}^{N} \frac{\beta_i}{\omega + \lambda_i}\right)\omega}\,. \qquad \text{(FNRP 5.53)}$$

This result is the **inhour equation** and represents a polynomial of degree $N+1$ whose roots are the ω_j.

4 The Reactor Period



Shown at the left are *traces* for several step insertions of reactivity. In

[1] Without delayed neutrons, the growth of a neutron population follows $e^{[(k-1)/l]t}$, so increasing l even a little slows that growth a lot!

all cases, after the immediate *jump* or *drop*, the solution follows

$$n(t) \approx A_1 e^{\omega_1 t} = A_1 e^{t/T},$$

where T is the **(stable) reactor period**.

Example: Estimate the period *graphically* for $\rho = 0.2\beta$, $\rho = \rho_A$, and $\rho = \rho_B$. Then, use the inhour equation to estimate ρ_A and ρ_B.

Ans.

5 Step Insertions with One Precursor Group

Write the PKEs as

$$\frac{d}{dt}\overbrace{\begin{bmatrix} n(t) \\ C(t) \end{bmatrix}}^{\mathbf{y}(t)} = \overbrace{\begin{bmatrix} (\rho-\beta)/\Lambda & \lambda \\ \beta/\Lambda & -\lambda \end{bmatrix}}^{\mathbf{M}} \begin{bmatrix} n(t) \\ C(t) \end{bmatrix}, \quad \begin{bmatrix} n(0) \\ C(0) \end{bmatrix} = \begin{bmatrix} n_0 \\ C_0 \end{bmatrix}.$$

The eigenvalues ω of **M** satisfy

$$\overbrace{1}^{a}\omega^2 + \overbrace{\left(\lambda - \frac{\rho-\beta}{\Lambda}\right)}^{b}\omega + \overbrace{\left(-\frac{\lambda\beta}{\Lambda} - \frac{\lambda(\rho-\beta)}{\Lambda}\right)}^{c} = 0 \longrightarrow$$

$$\boxed{\omega_{1/2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}.$$

The solutions are then

$$n(t) = A_1 e^{\omega_1 t} + A_2 e^{\omega_2 t} \qquad \text{and} \qquad C(t) = B_1 e^{\omega_1 t} + B_2 e^{\omega_2 t}.$$

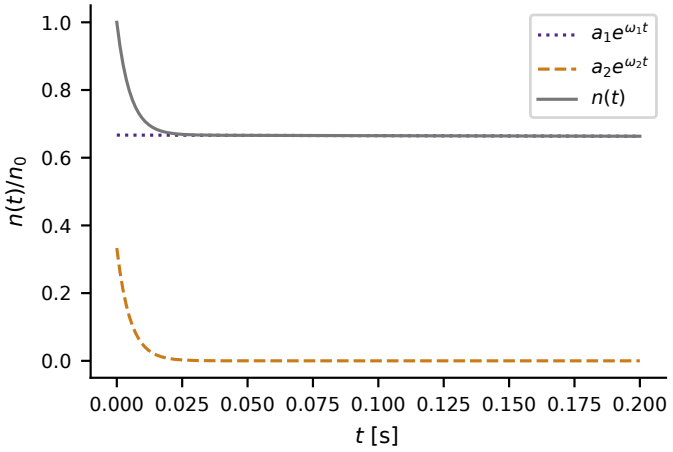
We need to find the four unknowns A_1, A_2, B_1, and B_2 noting that

- the initial conditions give two equations.
- the PKEs give two more equations, and setting $t = 0$ eliminates the exponentials!

6 Step Insertions with One Precursor Group

```
from math import sqrt
def step_insert(ρ, β, Λ, λ, n0):
  c0 = β*n0/(λ*Λ)
  a = 1
  b = λ*(1-(ρ-β)/(λ*Λ))
  c = -1*(λ*(ρ-β)/Λ + λ*β/Λ)
  d = sqrt(b**2-4*a*c)
  ω1 = (-b+d)/(2*a)
  ω2 = (-b-d)/(2*a)
  A1 = (ρ*n0/Λ - n0*ω2)/(ω1-ω2)
  A2 = n0 - A1
  B1 = (-β*n0*ω2/(λ*Λ))/(ω1-ω2)
  B2 = β*n0/(λ*Λ) - B1
  return ω1, ω2, A1, A2, B1, B2
```

$\rho = -0.5\beta$, $\Lambda = 5 \cdot 10^{-5}$



7 Prompt Jump/Drop Approximation

As the traces illustrate, after a small insertion $\Delta\rho \ll \beta$, the population swiftly jumps or drops. On the linear scale, the population appears to jump immediately from n_0 to some new value n_1. The ratio of this new value to the initial value satisfies

$$\boxed{\frac{n_1}{n_0} = \frac{\beta - \rho_0}{\beta - \rho_1}}, \qquad \text{(Prompt Jump Approximation)}$$

where ρ_0 is the initial reactivity (usually zero, but $|\rho_0| \ll \beta$ always) and the new reactivity $\rho_1 = \rho_0 + \Delta\rho$.

Example. Use the prompt jump approximation to verify the $\pm 0.2\beta$ traces. What is $\rho \pm 0.2\beta$ in \$? pcm?

8 Periods for Very Small and Very Large Insertions

A "small" reactivity is any $|\rho| \ll \beta$. In this case,

$$T \approx \frac{\beta}{\rho\lambda}. \qquad \text{(FNRP 5.57)}$$

For reactivities $\rho > \beta$, the neutron population grows exponentially with prompt neutrons alone and, hence, the period is very small and well approximated by

$$T \approx \frac{\Lambda}{\rho - \beta}. \qquad \text{(FNRP 5.58)}$$

Finally, for large *negative* insertions, $T \approx 1/\lambda$, or, for multiple precursor groups, $T \approx 1/\lambda_1$, where λ_1 corresponds to the longest lives group.